

NexSMS —Final Project Plan (Master Document)

Version: 1.0 (Final) **Status:** Approved —awaiting build start **Brand Name:** NexSMS **Target Audience:** Primarily Chinese users, international-friendly **Languages:** English + 中文 (toggle)

1. Product Overview

NexSMS is a Google Voice-style SMS/chat SaaS platform. Users get virtual phone numbers (10DLC assigned via providers like Twilio, SingleHouse, Plivo, Telnyx) and can send SMS to any number under their assigned number, with full chat-thread conversation history —exactly like Google Voice Messages.

Tagline (EN): Your Number. Your Conversation. **Tagline (中文):** 您的号码，您的对话

Core Concept

Admin buys numbers from Provider A/B/C → Number Pool
↓
User gets 1+ numbers assigned from pool (admin-managed)
↓
User sends SMS as that assigned number to any number
↓
Each contact = separate chat thread (Google Voice style)

2. Tech Stack

Layer	Technology
Frontend	React + Vite (SPA)
Backend	Node.js (Express)
Database	PostgreSQL
Queue/Jobs	Redis + BullMQ
SMS Providers	Twilio, SingleHouse, Plivo, Telnyx (pluggable)
Auth	JWT + OTP (2FA for admin)
i18n	English + 中文
Cache	Redis (theme, payment, settings)

2.1 CORE PRINCIPLE —Code-Free Admin Control (Admin-Code Editable Website-Configurable Admin Dashboard)

Admin-Code Editable Website-Configurable Admin Dashboard: Admin-Code Editable Website-Configurable Admin Dashboard — settings (DB) save, site-wide apply

Admin Dashboard Control (Code Editable):

Category	Admin-Code Editable Website-Configurable Admin Dashboard
Website Appearance	(primary/secondary), font, logo, favicon, homepage hero text, tagline, banner, footer text, button labels, 中文/English default

123-4567	[You] Hello! Your order is confirmed ☒	
+1 (555) 765-4321	[Client] Thank you! When will it arrive?	
	[Type a message... 	

- Left: inbox/thread list; Right: conversation
- Each thread header shows which assigned number is used
- Unread counts, message previews, timestamps
- “From number” selector per message (multi-number support)
- Keypad / “Enter a name or number” (as in reference screenshot)

3.3 Plans & Numbers Section (below chat)

- **Plan tab:** current plan, upgrade/downgrade
- **My Numbers tab:** all assigned numbers
 - Primary number badge
 - “+ Assign New Number” button (from pool)
 - Multiple numbers per user
 - Status (active/paused), switch primary, remove

3.4 Wallet

- Balance, transactions history
- Low-balance alert
- Auto cost deduction per SMS

3.5 Payment Page

- Shows only enabled gateways: Binance Pay, Bybit, Stripe, Cryptomus, TRC20 (USDT), BEP20 (USDT), BTC, BenlyPay
- Choose gateway → complete payment → wallet credited

3.6 Advanced User Features

- SMS Scheduling (schedule message for later)
- Bulk blast tool (with compliance/opt-in enforcement)
- AI Auto-reply (incoming → AI reply)
- Smart inbox (spam/sales filter)
- Reply suggestions
- Number Geo-Picker (filter by area code/country, if allowed)
- Analytics dashboard (volume, cost, delivery rate charts)
- User API access + API key management
- User webhooks (delivery callbacks to their own system)
- Message template library (OTP, notification, marketing)
- Referral / discount system (invite friend → credit bonus, admin-controlled rates)
- Multi-currency wallet (USD + USDT + regional, exchange rate admin-set)
- Public rate card page (transparent SMS rates per country)
- KYC verification (optional tier: unverified → limited limit, verified → larger limit)

4. SMS Engine (Core)

4.1 Multi-Provider Layer

- Pluggable common interface: `sendSms()`, `receiveWebhook()`, `checkNumber()`, `buyNumber()`
- Admin manages providers from dashboard (CRUD, credentials, enable/disable, priority)
- Provider health check + failover (auto or manual)
- Delivery status callbacks per provider (signature-verified)

4.2 Billing / Message Credits (2 modes, admin-controlled)

Billing Mode: ☐ Prepaid Credits (checkbox)
☐ Subscription Plan (checkbox)

- **Prepaid:** wallet balance; per-SMS cost deducted
- **Subscription:** monthly plan with SMS quota
- **Hybrid (optional):** quota exhausted → charge wallet, or block (admin rule)
- Per-user billing mode override by admin
- Super Admin: global mode toggle + rates control

4.3 Per-Number Daily Send Limit (24h window)

- Admin sets per-plan daily limit:
 - Starter → X SMS/day/number
 - Standard → Y SMS/day/number
 - Premier → Z SMS/day/number
- Separate limit for pay-per-SMS users (admin-controlled)
- Counter tracked in DB; resets after 24 hours
- On limit exceeded: block with “Try again in X hours”
- Admin panel shows per-number usage + reset timer

4.4 Delivery & Compliance

- Delivery reports: sent → delivered → failed + auto-retry
- Rate limiting per user (spam control)
- International routing intelligence (country → best/cheapest provider)
- A2P 10DLC compliance tracker (campaign status per number)
- MMS/attachment support (premium tier)

4.5 Fraud & Abuse Prevention (admin-toggable)

- **Flash USDT / fake crypto detection:**
 - Official contract address check only (TRC20/BEP20/BTC official contracts)
 - Blockchain explorer verification (TronGrid / BscScan / Blockchain.com) via TxID
 - Exact amount + receiver address match required
 - Min confirmations before credit (admin-set)
 - Test-transaction hold for large amounts (admin threshold)
- **Carding prevention (card gateways):**
 - Stripe Radar + 3D Secure
 - AVS/CVV verification
 - Velocity limit (same card/IP repeated attempts → auto-block)
 - Proxy/VPN detection → flag
 - Chargeback monitoring → alert + block
- **Risk scoring:** per-payment score (new user, large amount, VPN, new gateway → high risk flag)
- **Payment hold policy:** suspicious/large payments held → admin review → then credit
- **Admin review queue:** flagged payments require admin approval before credit
- **Replay protection + expiry:** order validated once; unpaid orders auto-cancel after 24h

- Per-number daily limits per plan + per user

5.7 Website Customization

- Color, font, logo, branding (DB-driven, instant apply)
- Full site theming from admin panel
- **Site Content Editor:** homepage hero text, tagline, features section text, FAQ, pricing text, footer, notification banner — ☐ admin-☐ edit ☐ ☐ ☐ ☐
- **Translation Editor:** English + 中文 ☐ ☐ text admin panel ☐ ☐ ☐ edit (☐ ☐ ☐ code ☐ ☐)

5.7.1 System Settings (Admin General Controls)

- Site name, logo, favicon, default language
- Maintenance mode (ON/OFF + message)
- Backup/export buttons
- Alert notification settings

5.8 Feature Toggles (Admin Control Hub)

- **Settings → Feature Toggles** page: ☐ ☐ ☐ ☐ ☐ ☐ feature-☐ ☐ ON/OFF switch
- KYC required (ON/OFF + threshold)
- Referral system (ON/OFF + bonus rate)
- Multi-currency wallet (ON/OFF + exchange rates)
- Fraud protection modules (ON/OFF individually): flash-USDT check, carding check, risk scoring, payment hold, IP blocklist, message content filter
- Number expiry/renewal (ON/OFF + grace period)
- Maintenance mode (ON/OFF + message)
- Email/SMS alerts (ON/OFF)
- Bulk blast (ON/OFF)
- AI features (ON/OFF)
- User API access (ON/OFF)
- All toggles apply instantly, site-wide

5.9 Security & Ops

- Webhook manager + signature validation
 - 2FA (OTP login) + IP whitelisting for admin
 - Audit log (every admin action)
 - Backup / export (CSV/JSON)
 - Platform-wide analytics + provider cost report
 - Admin alert center (provider down, big payment, low balance, suspicious activity)
-

6. Database Model (概要)

- users / admins
- providers (type, credentials, active, priority)
- numbers (number, provider_id, assigned_user_id, status, geo)
- assignments (user ↔ number, primary flag)
- conversations (user_id, assigned_number_id, contact_number)
- messages (conversation_id, direction, body, status, provider_id, cost)
- wallets / transactions
- plans / subscriptions
- daily_limits (number_id, date, count)
- settings (billing mode, theme, branding, i18n)

-

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

CNC

[illegible][illegible][illegible]

M. J. Griffin, D. L. Brown, A. C. Knight

6421

[illegible]

1. 1910 2. 1911 3. 1912 4. 1913 5. 1914 6. 1915 7. 1916 8. 1917 9. 1918 10. 1919 11. 1920 12. 1921 13. 1922 14. 1923 15. 1924 16. 1925 17. 1926 18. 1927 19. 1928 20. 1929 21. 1930 22. 1931 23. 1932 24. 1933 25. 1934 26. 1935 27. 1936 28. 1937 29. 1938 30. 1939 31. 1940 32. 1941 33. 1942 34. 1943 35. 1944 36. 1945 37. 1946 38. 1947 39. 1948 40. 1949 41. 1950 42. 1951 43. 1952 44. 1953 45. 1954 46. 1955 47. 1956 48. 1957 49. 1958 50. 1959 51. 1960 52. 1961 53. 1962 54. 1963 55. 1964 56. 1965 57. 1966 58. 1967 59. 1968 60. 1969 61. 1970 62. 1971 63. 1972 64. 1973 65. 1974 66. 1975 67. 1976 68. 1977 69. 1978 70. 1979 71. 1980 72. 1981 73. 1982 74. 1983 75. 1984 76. 1985 77. 1986 78. 1987 79. 1988 80. 1989 81. 1990 82. 1991 83. 1992 84. 1993 85. 1994 86. 1995 87. 1996 88. 1997 89. 1998 90. 1999 91. 2000 92. 2001 93. 2002 94. 2003 95. 2004 96. 2005 97. 2006 98. 2007 99. 2008 100. 2009 101. 2010 102. 2011 103. 2012 104. 2013 105. 2014 106. 2015 107. 2016 108. 2017 109. 2018 110. 2019 111. 2020 112. 2021 113. 2022 114. 2023 115. 2024 116. 2025 117. 2026 118. 2027 119. 2028 120. 2029 121. 2030 122. 2031 123. 2032 124. 2033 125. 2034 126. 2035 127. 2036 128. 2037 129. 2038 130. 2039 131. 2040 132. 2041 133. 2042 134. 2043 135. 2044 136. 2045 137. 2046 138. 2047 139. 2048 140. 2049 141. 2050 142. 2051 143. 2052 144. 2053 145. 2054 146. 2055 147. 2056 148. 2057 149. 2058 150. 2059 151. 2060 152. 2061 153. 2062 154. 2063 155. 2064 156. 2065 157. 2066 158. 2067 159. 2068 160. 2069 161. 2070 162. 2071 163. 2072 164. 2073 165. 2074 166. 2075 167. 2076 168. 2077 169. 2078 170. 2079 171. 2080 172. 2081 173. 2082 174. 2083 175. 2084 176. 2085 177. 2086 178. 2087 179. 2088 180. 2089 181. 2090 182. 2091 183. 2092 184. 2093 185. 2094 186. 2095 187. 2096 188. 2097 189. 2098 190. 2099 191. 2100 192. 2101 193. 2102 194. 2103 195. 2104 196. 2105 197. 2106 198. 2107 199. 2108 200. 2109 201. 2110 202. 2111 203. 2112 204. 2113 205. 2114 206. 2115 207. 2116 208. 2117 209. 2118 210. 2119 211. 2120 212. 2121 213. 2122 214. 2123 215. 2124 216. 2125 217. 2126 218. 2127 219. 2128 220. 2129 221. 2130 222. 2131 223. 2132 224. 2133 225. 2134 226. 2135 227. 2136 228. 2137 229. 2138 230. 2139 231. 2140 232. 2141 233. 2142 234. 2143 235. 2144 236. 2145 237. 2146 238. 2147 239. 2148 240. 2149 241. 2150 242. 2151 243. 2152 244. 2153 245. 2154 246. 2155 247. 2156 248. 2157 249. 2158 250. 2159 251. 2160 252. 2161 253. 2162 254. 2163 255. 2164 256. 2165 257. 2166 258. 2167 259. 2168 260. 2169 261. 2170 262. 2171 263. 2172 264. 2173 265. 2174 266. 2175 267. 2176 268. 2177 269. 2178 270. 2179 271. 2180 272. 2181 273. 2182 274. 2183 275. 2184 276. 2185 277. 2186 278. 2187 279. 2188 280. 2189 281. 2190 282. 2191 283. 2192 284. 2193 285. 2194 286. 2195 287. 2196 288. 2197 289. 2198 289. 2199 290. 2200 291. 2201 292. 2202 293. 2203 294. 2204 295. 2205 296. 2206 297. 2207 298. 2208 299. 2209 2210. 2211. 2212. 2213. 2214. 2215. 2216. 2217. 2218. 2219. 2220. 2221. 2222. 2223. 2224. 2225. 2226. 2227. 2228. 2229. 2230. 2231. 2232. 2233. 2234. 2235. 2236. 2237. 2238. 2239. 2240. 2241. 2242. 2243. 2244. 2245. 2246. 2247. 2248. 2249. 2250. 2251. 2252. 2253. 2254. 2255. 2256. 2257. 2258. 2259. 2260. 2261. 2262. 2263. 2264. 2265. 2266. 2267. 2268. 2269. 2270. 2271. 2272. 2273. 2274. 2275. 2276. 2277. 2278. 2279. 2280. 2281. 2282. 2283. 2284. 2285. 2286. 2287. 2288. 2289. 2290. 2291. 2292. 2293. 2294. 2295. 2296. 2297. 2298. 2299. 2300. 2301. 2302. 2303. 2304. 2305. 2306. 2307. 2308. 2309. 2310. 2311. 2312. 2313. 2314. 2315. 2316. 2317. 2318. 2319. 2320. 2321. 2322. 2323. 2324. 2325. 2326. 2327. 2328. 2329. 2330. 2331. 2332. 2333. 2334. 2335. 2336. 2337. 2338. 2339. 2340. 2341. 2342. 2343. 2344. 2345. 2346. 2347. 2348. 2349. 2350. 2351. 2352. 2353. 2354. 2355. 2356. 2357. 2358. 2359. 2360. 2361. 2362. 2363. 2364. 2365. 2366. 2367. 2368. 2369. 2370. 2371. 2372. 2373. 2374. 2375. 2376. 2377. 2378. 2379. 2380. 2381. 2382. 2383. 2384. 2385. 2386. 2387. 2388. 2389. 2390. 2391. 2392. 2393. 2394. 2395. 2396. 2397. 2398. 2399. 2400. 2401. 2402. 2403. 2404. 2405. 2406. 2407. 2408. 2409. 24

... ..

- $\text{E} \times (\alpha | \rho | \beta | \gamma | \delta)$

- ...if we have a ...

- [illegible]

- ...is controlled (see my personal subscription, both

- unisolated single-provider outages

- ☐
- ☐
- ☐
- configurable
- ☐
- ☐
- ☐
- ☐
- ☐
- ☐
- admin dashboard
- ☐
- ☐
- ☐

- configurable admin dashboard